Degeneracy of the Inverse Problem: Uncountable Solutions from Dynamics

1 The Neural Model

The flyvis graded-voltage ODE for each postsynaptic neuron i is

$$\tau_i \frac{dv_i(t)}{dt} = -v_i(t) + V_i^{\mathrm{rest}} + \sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij}\,\mathrm{ReLU}\big(v_j(t)\big) + I_i(t) \tag{1}$$

where τ_i and V_i^{rest} are cell-type parameters, $\mathbf{W}_{ij}$ is the connectome-constrained synaptic weight from neuron j to i, and $I_i(t)$ is the external stimulus. **The inverse problem**: given observed trajectories $\{v_i(t)\}_{t=0}^{T}$, stimuli $\{I_i(t)\}$, and the binary adjacency matrix $\mathbf{A}$ (which edges exist), recover the connectivity weights $\mathbf{W}$, time constants τ, and resting potentials V^{rest}.

2 The Degeneracy

2.1 Rearranging to a linear system per neuron

Rearranging Eq. **??** to isolate the incoming synaptic weights, we move them to the left-hand side:

$$\sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij}\,\mathrm{ReLU}\big(v_j(t)\big) = \tau_i \frac{dv_i(t)}{dt} + v_i(t) - V_i^{\mathrm{rest}} - I_i(t) \tag{2}$$

We define the presynaptic activity $h_j(t) = \mathrm{ReLU}\big(v_j(t)\big)$ and the residual $b_i(t) = \tau_i \frac{dv_i(t)}{dt} + v_i(t) - V_i^{\mathrm{rest}} - I_i(t)$. The constraint is then:

$$\sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij}\, h_j(t) = b_i(t) \qquad \forall\, t \in \{1, \ldots, T\} \tag{3}$$

We stack these T timepoint constraints into a linear system: Let $\mathbf{H}_i \in \mathbb{R}^{T \times d_i}$ be the activity matrix (rows = time, columns = the d_i presynaptic neurons), $\mathbf{w}_i \in \mathbb{R}^{d_i}$ the incoming weight vector, and $\mathbf{b}_i \in \mathbb{R}^T$ the stacked residuals. Then:

$$\mathbf{H}_i\,\mathbf{w}_i = \mathbf{b}_i \tag{4}$$

This linear equation holds for each postsynaptic neuron $i = 1, \ldots, N$, where $N = 13{,}697$ and d_i ranges from 1 to 208 (the in-degree distribution of the flyvis connectome).

2.2 Global SVD approach (coarse bound)

Perturbations $\boldsymbol{\delta} \in \ker(\mathbf{H}_i)$ satisfy $\mathbf{H}_i \boldsymbol{\delta} = \mathbf{0}$, preserving dynamics. By rank-nullity:

$$d_i = \mathrm{rank}(\mathbf{H}_i) + \dim\ker(\mathbf{H}_i) \tag{5}$$

We compute the effective rank of the population activity matrix via SVD of the subsampled activity $\mathbf{H} \in \mathbb{R}^{8000 \times 982}$ (64,000 frames subsampled to 8,000; 13,741 neurons subsampled to

982). The effective rank at 99% cumulative variance is 45. Since each neuron's activity $\mathbf{H}_i$ is a submatrix of the global activity, we have $\operatorname{rank}(\mathbf{H}_i) \leq 45$, giving:

$$\dim\ker(\mathbf{H}_i) \approx \max(0,\ d_i - 45) \tag{6}$$

Summing over all neurons:

$$\text{total null dim} = \sum_{i=1}^{13,697} \max(0,\ d_i - 45) = 115,223 \quad (26.5\% \text{ of edges}) \tag{7}$$

We report sensitivity to the variance threshold.

Variance threshold	Effective rank r	Null space dim	% unconstrained
90%	1	420,415	96.8%
95%	2	407,427	93.9%
99%	**45**	**115,223**	**26.5%**
99.5%	90	26,413	6.1%
99.9%	238	0	0.0%

2.3 Per-neuron SVD approach (empirical measurement)

While the global SVD provides a population-level upper bound, we reveal which neurons are identifiable and which are degenerate by computing the SVD of $\mathbf{H}_i$ separately for each postsynaptic neuron i. For each neuron, we find its effective rank r_i at various variance thresholds and apply rank-nullity:

$$\dim(\ker(\mathbf{H}_i)) \approx d_i - r_i \tag{8}$$

where r_i is the smallest integer such that the top r_i singular values of $\mathbf{H}_i$ explain at least θ fraction of its variance. A neuron is **fully identifiable** if $d_i \leq r_i$ (all incoming weights constrained by dynamics) and **degenerate** if $d_i > r_i$ (some weights free to vary). We compute per-neuron SVD using training data (x_list_train/voltage.zarr, 64,000 frames subsampled to 8,000). For each postsynaptic neuron i with $d_i \geq 1$ incoming edges, we extract the $T \times d_i$ activity matrix $\mathbf{H}_i$, compute its SVD, and find the effective rank at multiple variance thresholds. Per-neuron SVD reveals substantially stronger degeneracy than the global bound: 80.3% of edges are unconstrained at 99% (vs. 26.5% global), reflecting strong individual neuron-level input correlations from same-type inputs across columns.

Variance threshold	Mean rank $\bar{r}_i$	Null space dim	% unconstrained
90%	1.2	417,563	96.2%
95%	2.0	406,901	93.7%
99%	**6.2**	**348,545**	**80.3%**
99.5%	8.9	312,396	72.0%
99.9%	15.4	223,775	51.5%

2.4 Within-type degeneracy and structural counting

The columnar organization of the connectome forces same-type neurons to have nearly identical activity. The flyvis circuit spans 217 columns, each covering a position in the visual field. When a moving stimulus crosses the eye, column i sees what column $i-1$ saw moments earlier—so same-type neurons across columns produce time-shifted copies of the same signal. In the activity matrix $\mathbf{H}_i$, when a postsynaptic neuron receives input from k copies of the same cell type, these k columns are strongly correlated, carrying rank ≈ 1 instead of k, and contributing

$k - 1$ dimensions to $\ker(\mathbf{H}_i)$. When the k columns are identical ($\mathbf{h}_{j_1} = \cdots = \mathbf{h}_{j_k} = \mathbf{h}$), a null perturbation $\mathbf{H}_i\boldsymbol{\delta} = \mathbf{0}$ requires $(\sum_{m=1}^{k} \delta_{j_m})\,\mathbf{h} = \mathbf{0}$, so $\sum_{m=1}^{k} \delta_{j_m} = 0$—a sum-zero constraint leaving $k - 1$ free directions. Summing over all neurons and cell types:

$$\dim \ker(\mathbf{H}) = \sum_{i=1}^{N} \sum_{\alpha:\, k_{i\alpha}>1} (k_{i\alpha} - 1) \tag{9}$$

For the flyvis connectome (434,112 edges, 65 cell types), this yields a null space dimension of $\sim$ 121,100 (approximately 28% of all edges). Since each null direction scales continuously as $\boldsymbol{\delta} \to \lambda\boldsymbol{\delta}$, the solution set is an **uncountably infinite** $\sim$ 121,100-dimensional manifold. This structural count agrees to within 5% with the global SVD bound (115,223), validating both approaches and confirming that within-type degeneracy dominates.

3 Empirical Verification

The null space estimate relies on the effective rank at 99% variance—a conventional threshold, not an exact quantity. To test whether the predicted null directions are real, we verify empirically: if the null space analysis is correct, then a perturbed matrix $\mathbf{W} + \boldsymbol{\delta}$ with low connectivity R^2 (weights changed substantially) should still produce high rollout R^2 (dynamics nearly identical).

3.1 Non-degenerate types: structural constraints

Of 65 cell types, 13 lack degenerate groups: R1, L3, Lawf2, Mi1, Mi4, Mi9, Mi10, Mi11, Mi12, Mi13, T1, Tm5b, Tm9. These exhibit strict one-to-one connectivity patterns. Early sensory types (R1, L3) and medulla intrinsic cells (Mi1, Mi4, Mi9-13) show this constraint, suggesting fixed connectivity preserves retinotopic spatial information. These 13 types are excluded from variant generation.

3.2 Variant generation method

In contrast, 52 cell types with degenerate groups support flexible computation via weight redistribution. For each postsynaptic neuron i, we count how many presynaptic neurons of the same cell type project to it. If $k > 1$, those k edges form a degenerate group. We generate a sum-preserving perturbation $\boldsymbol{\delta}$ (with $\sum_j \delta_j = 0$) and apply it at 15 logarithmically-spaced scales ($\lambda = 0.05$ to 8.0), yielding 960 single-type variants. For mixed-type variants, we select the top 10 cell types by total null space dimension and simultaneously perturb all 10 with independent perturbations at random amplitudes, generating 1,000 more variants.

3.3 Results

For each variant, we compute connectivity R^2 (weight preservation) and rollout R^2 (ODE trajectory preservation via 1,000-frame simulation). Single-type variants show mean connectivity $R^2 = 0.9802 \pm 0.0761$ and rollout $R^2 = 0.9965 \pm 0.0447$; mixed-type variants show 0.8387 ± 0.1644 and 0.9973 ± 0.0037 respectively. The striking pattern is the decoupling: mixed-type rollout stays near-perfect despite substantially lower connectivity R^2 (seeF igure **??**). Table **??** focuses on the limit case ($\lambda = 8.0$) across all 51 degenerate types. Color-coding reveals connectivity strength: green ($R^2 > 0.95$) indicates modest perturbations, orange ($0.50 < R^2 \leq 0.95$) moderate, and red ($R^2 \leq 0.50$) severe. Tm5c achieves the lowest connectivity R^2 (0.2823, red) while maintaining rollout $R^2 = 0.9965$. In contrast, C3 and Tm5Y show perfect preservation ($R^2 = 1.0$ both), indicating minimal degenerate structure.

Cell Type	Conn R^2	Rollout R^2						
R2	0.9421	1.0000	CT1(M10)	0.8774	0.9989	Tm4	0.9945	1.0000
R3	0.9672	1.0000	Mi2	0.9146	0.9809	Tm5Y	1.0000	1.0000
R4	0.7408	0.9974	Mi3	0.9375	0.9984	Tm5a	0.9694	0.9996
R5	0.9921	0.9999	Mi14	0.9649	0.9999	Tm5c	0.2823	0.9965
R6	0.9995	1.0000	Mi15	0.8710	0.9994	Tm16	0.9899	1.0000
R7	0.9987	1.0000	T2	0.9692	0.9922	Tm20	0.9966	1.0000
R8	0.8993	0.9998	T2a	0.9688	0.9998	Tm28	0.7526	0.9869
L1	0.7275	0.9979	T3	0.9664	0.9996	Tm30	0.5658	0.9982
L2	0.6171	0.9980	T4a	0.9668	0.9983	TmY3	0.9983	1.0000
L4	0.9961	0.9983	T4b	0.9904	0.9985	TmY4	0.9931	1.0000
L5	0.7332	0.9944	T4c	0.9875	0.9990	TmY5a	0.9409	0.9999
Lawf1	0.9967	1.0000	T4d	0.9778	0.9988	TmY9	0.9517	0.9992
Am	0.9512	0.9999	T5a	0.9975	0.9999	TmY10	0.9723	0.9616
C2	0.9270	0.9996	T5b	0.9977	1.0000	TmY13	0.9942	1.0000

Table 1: Single-type variants at extreme perturbation scale $\lambda = 8.0$ for 51 degenerate cell types. Connectivity R^2 (Conn R^2) measures weight preservation; rollout R^2 (Rollout R^2) measures dynamics preservation. Color-coding: green ($R^2 > 0.95$), orange ($0.50 < R^2 \leq 0.95$), red ($R^2 \leq 0.50$).

4 Discussion

4.1 The inverse problem is fundamentally ill-posed

Recovering the connectivity matrix $\mathbf{W}$ from dynamics alone is ill-posed. The null space analysis quantifies the ill-posedness: the weight space has ~121,100 **null dimensions** out of 434,112 edges (~28%), along which weights can be redistributed within same-type groups without affecting the dynamics. The empirical verification confirms this across all 52 non-retina cell types: at $\lambda = 8.0$, connectivity R^2 drops as low as 0.28 while rollout R^2 stays above 0.96. The set of valid solutions is **uncountably infinite**: the solution manifold is a ~121,100-dimensional continuous subspace of $\mathbb{R}^{434,112}$. The ill-posedness is **structural**, arising from the columnar organization of the circuit: same-type neurons in different columns produce strongly correlated activity, making their individual contributions to a shared target indistinguishable from dynamics alone.

4.2 Implications for GNN-based inference

The GNN approximates the dynamics as:

$$\frac{\widehat{dv_i(t)}}{dt} = f_\theta\Big(v_i(t), \mathbf{a}_i, \sum_{j \in \mathcal{N}_i} \widehat{\mathbf{W}}_{ij}\, g_\phi\big(v_j(t), \mathbf{a}_j\big)^2, I_i(t)\Big) \tag{10}$$

where $f_\theta = \mathrm{MLP}_0$ and $g_\phi = \mathrm{MLP}_1$ are three-layer perceptrons, $\widehat{\mathbf{W}}_{ij}$ are learnable edge weights, and $\mathbf{a}_i \in \mathbb{R}^2$ are learnable per-neuron embeddings. The training minimizes:

$$\mathcal{L}_{\text{pred}} = \sum_{i,t} \big\|\widehat{y}_i(t) - y_i(t)\big\|_2 \tag{11}$$

between simulator $y_i(t) = dv_i(t)/dt$ and GNN predictions $\widehat{y}_i(t) = \widehat{dv_i}(t)/dt$, augmented with regularization:

$$\begin{aligned} \mathcal{L} = {} & \|\widehat{\mathbf{y}} - \mathbf{y}\|_2 + \lambda_0\|\theta\|_1 + \lambda_1\|\phi\|_1 + \lambda_2\|\widehat{\mathbf{W}}\|_1 \\ & + \gamma_0\|\theta\|_2 + \gamma_1\|\phi\|_2 \\ & + \mu_0 \,\big\|\, \mathrm{ReLU}\big(-\tfrac{\partial g_\phi(v,\mathbf{a})}{\partial v}\big)\big\|_2 + \mu_1 \,\|g_\phi(v_\star, \mathbf{a}) - v_\star\|_2 \end{aligned} \tag{12}$$

where λ terms promote sparsity, γ terms stabilize learned functions, μ_0 enforces monotonicity of the edge message, and μ_1 normalizes with respect to reference voltage $v_\star$. When the GNN achieves $R^2_{\mathbf{W}} = 0.997$ (as our LLM-optimized model does at $\sigma = 0.5$), it is recovering the $\sim$313,000 identifiable parameters well while the $\sim$121,100 null-space parameters are essentially free. The GNN's implicit inductive bias (message-passing, weight sharing) acts as an implicit regularizer that selects a particular solution from the uncountable solution manifold.

4.3 Why noise helps

Adding process noise $\sigma\eta_i(t)$ to the simulation:

$$\tau_i \frac{dv_i(t)}{dt} = -v_i(t) + V_i^{\text{rest}} + \sum_{j \in \mathcal{N}_i} \mathbf{W}_{ij}\, \mathrm{ReLU}\big(v_j(t)\big) + I_i(t) + \sigma\eta_i(t) \tag{13}$$

breaks the within-type degeneracy. Each neuron receives independent noise realizations, so even same-type neurons now produce distinguishable signals. This independent variation perturbs the activity matrix $\mathbf{H}_i$, breaking the strong within-type correlations that cause rank deficiency. As a result, the effective rank of $\mathbf{H}_i$ increases and the null space shrinks, making more edge weights identifiable by the SVD criterion $\mathrm{rank}(\mathbf{H}_i) \geq d_i$. We apply the same global SVD approach described in §**??** to activity matrices computed at three noise levels ($\sigma = 0$, 0.05, 0.5). For each level, we extract the effective rank r at 99% variance and compute the null space dimension using the per-neuron formula. The $\sigma = 0$ row reproduces our earlier global SVD estimate and validates the methodology; the noisy conditions use identical subsampling for direct comparison.

Noise	Activity rank r	Null space dim	% identifiable	GNN $R^2_{\mathbf{W}}$
$\sigma = 0$	45	115,223	73.5%	0.899 ± 0.030
$\sigma = 0.05$	134	10,227	97.6%	0.962 ± 0.012
$\sigma = 0.5$	**781**	**0**	**100.0%**	$\mathbf{0.997 \pm 0.000}$

The trend is monotonic: noise increases the effective rank ($45 \to 134 \to 781$), shrinking the null space and improving identifiability. At $\sigma = 0.5$, the rank (781) far exceeds the maximum in-degree (208), so every neuron's weights are fully identifiable—the null space vanishes and $R^2_{\mathbf{W}} = 0.997$. The two noisy conditions show strong agreement between theoretical identifiability and observed recovery (97.6% identifiable $\to$ 0.962 observed, 100% $\to$ 0.997). The noise-free case (0.899 observed vs. 73.5% identifiable) exceeds the bound, likely because GNN regularization biases null-space weights toward correlated ground-truth values. To break the within-type correlations experimentally, optogenetics could inject independent perturbations into each neuron of a given type (e.g., stimulate each L1 with a unique light pattern), or selectively ablate same-type neurons across columns (e.g., silence L1 in columns 4 and 6 but not 5). Both approaches would increase $\mathrm{rank}(\mathbf{H}_i)$ and shrink the null space—the same mechanism as noise, but achievable in real tissue.

5 Summary

Recovering the connectome from neural dynamics is fundamentally ill-posed: the columnar organization of the flyvis circuit forces same-type neurons across columns to produce strongly correlated activity, making their individual synaptic contributions indistinguishable. Two independent estimates of the null space—a global SVD bound (115,223 dimensions) and a per-type structural count ($\sim$121,100 dimensions)—agree to within 5%. The global bound captures all correlations (within- and cross-type) but is coarse, while the per-type count precisely measures within-type redundancy but misses cross-type correlations. This structural degeneracy affects about 25% of all 434,112 edge weights. This is supported empirically by generating 780 perturbed connectivity matrices where weights change measurably (conn. R^2 ranging from 0.28 to 1.0, most between 0.93 and 0.99) while dynamics remain nearly identical (rollout $R^2 > 0.96$). Adding independent process noise breaks the within-type correlations and increases the effective activity rank from 45 (noise-free) to 781 ($\sigma = 0.5$), eliminating the null space entirely and enabling the GNN to achieve $R^2_{\mathbf{W}} = 0.997$. The entire analysis rests on the rank-nullity theorem: a neuron's incoming weights are recoverable when the activity rank exceeds its in-degree. Since the per-neuron rank cannot be measured directly, we upper-bound it by the global effective rank of the population activity (computed once by SVD); the only approximation is choosing the variance threshold that defines this effective rank. **The condition $r_{\mathbf{global}} \geq d_i$ provides a principled, per-neuron criterion for when connectome recovery from dynamics is well-posed or ill-posed.**

scatterplot_r2_variants.png

Figure 1: Connectivity R^2 versus rollout R^2 for 1,960 degenerate connectivity variants. Single-type: 957 variants across 52 cell types at 15 perturbation scales. Mixed-type: 1,000 variants perturbing top 10 types simultaneously.